

An Early Developmental Marker of Deficit versus Nondeficit Schizophrenia

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Abstract

Background: People with schizophrenia who have primary negative symptoms (deficit schizophrenia) differ from those without such symptoms (nondeficit schizophrenia) relative to risk factors, course of illness, other signs and symptoms, treatment response, and biological correlates. Given the evidence for gestational and perinatal risk factors for schizophrenia, the deficit/nondeficit differences suggest these two groups have developmental differences as well. We previously reported a difference between people with schizophrenia and comparison subjects in the width of the palate. The palate attains its postnatal shape by gestational week 17. We tested the hypothesis that people with deficit and nondeficit schizophrenia would differ on palate width.

Methods: A dentist made blind assessments of the width, depth, and length of the palate in deficit (N=21) and nondeficit (N=25) patients and control subjects (N=127) matched for age and gender.

Results: Deficit patients had wider palates than both nondeficit patients ($P<0.0001$) and control subjects ($p<0.0001$), with respective effect sizes of 1.08 and 1.01. There were no significant deficit/nondeficit differences in palate length or depth, and no difference between nondeficit and comparison subjects in width, length, or depth.

Conclusions: Palate width reflects a divergence in development between deficit and nondeficit patients, which may occur by the early second trimester in some deficit patients. This difference is consistent with the hypothesis that deficit schizophrenia is a separate disease within the syndrome of schizophrenia. It may also point to a risk factor specific to deficit rather than nondeficit schizophrenia.

Key words:

pregnancy

negative symptoms

palate

delayed effects, prenatal exposure (MESH term)

cytomegalovirus

toxoplasmosis

1. Introduction

Schizophrenia has a great deal of heterogeneity in risk factors, signs and symptoms, course of illness, etiological/risk factors, and treatment response. This variation raises the question of whether schizophrenia is a syndrome rather than a single disease, analogous to, for instance, dementia, which can be caused by more than one disease.

One approach to understanding this heterogeneity has been study of people with primary negative symptoms. People with schizophrenia who have these features meet criteria for deficit schizophrenia, while people without these features have nondeficit schizophrenia (Kirkpatrick et al., 1989). As the deficit and nondeficit groups differ relative to the dimensions used to distinguish diseases—signs and symptoms, risk factors, course of illness, etiological/risk factors, and treatment response—we proposed that deficit schizophrenia is a separate disease within the syndrome of schizophrenia (Kirkpatrick et al., 2001). Subsequent research has provided additional evidence supporting this concept (Kirkpatrick & Galderisi, 2008; Kirkpatrick et al., 2017). Given the extensive evidence on prenatal and childhood/adolescent risk factors for schizophrenia (Matheson et al., 2011; Cannon et al., 2002; Geddes et al., 1999), one could expect that there would be developmental differences between deficit and nondeficit schizophrenia. One developmental risk factor for deficit but not nondeficit schizophrenia—an increase in summer rather than winter birth—has been replicated several times (Messias et al., 2004).

Minor physical anomalies (MPA), which have an increased prevalence in schizophrenia, are theoretically interesting because the gestational time period in which each one develops is usually known (Compton et al., 2011; Franco et al., 2010). An abnormal palate is part of the Waldrop physical anomaly scale, which has been used extensively in study of MPAs in schizophrenia and other disorders, but little attention has been paid to this single item. That scale rates for the presence (abnormal) or absence of a high or steeped palate. A blinded study using casts of the palate and teeth found an increase in width but not depth or length of the palate in people with schizophrenia, with a large effect size (Kirkpatrick et al., 2007). A second blinded study (Delice et al., 2016) of palate casts from people with schizophrenia and comparison subjects also found an increase in palate width.

We hypothesized that an abnormally wide palate would be found in deficit but not nondeficit patients and tested this hypothesis in samples from the study of Delice et al. (2016).

2. Methods and Materials

2.1 Subjects:

Subjects and data came from the study of Delice et al. (2016), and further details of the sample and measurement methods can be found in the original report. In brief, they made casts of the palate of their subjects, then measured palate width, length, and depth blind to group membership (schizophrenia vs. comparison subjects) using landmarks as described (Delice et al., 2016). In that study, 56 people in that study who had schizophrenia had ratings on the Positive and Negative Syndromes Scale (PANSS) and were included in the present study, along with 127 comparison subjects without schizophrenia matched to the patients on age and gender. All of the subjects were Turkish.

2.2 Deficit/nondeficit categorization

We used the Proxy for the Deficit Syndrome to distinguish deficit vs. nondeficit patients, as used previously (for instance, Garcia-Rizo et al., 2012; Kirkpatrick et al., 1993, Messias et al., 2001). This approach has been validated by comparison categorizations based on the Schedule for the Deficit Syndrome, the standard instrument for the diagnosis of deficit and nondeficit patients (Kirkpatrick et al., 1989, 1993; Messias et al., 2001). The PDS quantifies the combination of high